

Linux-based SeaBED AUV measures Antarctic ice thickness

The Linux-based SeaBED AUV (autonomous underwater vehicle) has been used by Woods Hole Oceanographic Institute (WHOI) to create the first ever 3D map of the Antarctic sea ice. The final outcome shows that the ice is thicker than earlier estimated.

Massachusetts-based WHOI announced that it has published a paper in Nature Geoscience on its experiments which have been conducted by SeaBED, an autonomous, Linux-based submarine. SeaBED performed an underwater survey



which indicated that Antarctic sea ice is actually thicker than previous estimates revealed. With this AUV, WHOI has created the world's first high-resolution 3D map of the Antarctic sea ice.

Previous estimates were based on satellite imaging and some other resources which concluded that the ice was much thinner.

Global warming is melting the Arctic ice fairly fast, so the greater volume of ice locked up in the Antarctic Sea could have a balancing effect on rising sea-levels.

Hanumant Singh, an engineering scientist at WHOI, designed, built and operated the AUV at his lab, and he said it's the device's 'manoeuvrability and stability' that has enabled it to be deployed in such challenging situations. SeaBED is about two metres long and weighs almost 200 kilograms. It runs on Ubuntu Linux with a 1.2GHz Intel Pentium chip. Its entire source code is shared with users.

CoreOS introduces Rocket, its own open source container technology

CoreOS, which is a start-up in the server operating system space, is no more willing to remain overshadowed by enterprise giants like Red Hat and aims to create its own identity now. CoreOS will no more be limited to Docker and the trendy container technology, but will set its own standards with an app container.

This news has been confirmed in a blog post by CoreOS founder and chief executive Alex Polvi. Docker's security model and complex platform have become a huge concern for CoreOS, and that's why the start-up has developed Rocket as an alternative to the open source container technology. The Rocket container runtime was released recently, and several issues that Linux distributions faced with Docker have been resolved. CoreOS had been an early supporter of Docker.

Rocket is a simple command line program to run an app container image or ACI. It gets encrypted and distributed across multiple channels. Rocket can be used not only on CoreOS but on any Linux distribution. CoreOS will



OpenSUSE gets live kernel patching!



SUSE recently announced the availability of SUSE Linux Enterprise Live Patching. SUSE is using open source kGraft for the live patching. kGraft has a GPLv2 licence. OpenSUSE users now will be able to perform critical updates without rebooting the system.

Live kernel patching is being pushed in the upstream Linux kernel. Hopefully, all Linux distributions will adopt to it in the near future. kGraft only patches the new updated part of packages and pushes the Linux kernel to continue functioning even during the updating process.

SUSE's product manager, Matthias Eckermann, said, "Our goal is to increase the service availability even when SUSE is being updated. We want to reduce the system downtime by patching the updates frequently. SUSE Linux Enterprise Live Patching will improve security and stability by applying interval patches. It is an open source solution that offers zero-interruption interactions with the system. Live kernel patching is the ideal solution for mission-critical systems, databases, and extended simulations which need maximum uptime."

As of now, SUSE has deployed the live kernel patching feature for SUSE Linux Enterprise Server (SLES) 12 x86_64 bit version. SUSE users with the SLES 12 Priority Support Subscription can use the feature by taking help from SUSE service engineers. This feature will definitely benefit servers running SLES by reducing downtime to a great extent.